

Roll No.

TEE-303

B. TECH. (EE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

ELECTRICAL AND ELECTRONIC MEASURING INSTRUMENTS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss accuracy, sensitivity, reproducibility, drift, static error and dead zone. Also explain how are they related. (CO1)

OR

- (b) Discuss absolute instrument, secondary instrument, analog instrument, digital instrument, null type instrument and deflection type instrument. (CO1)

2. (a) A 470 ohm $\pm 10\%$ resistor has a potential difference of 12 V across its terminals. If the voltage is measured with an accuracy of $\pm 6\%$, determine the power dissipated in the resistor and specify the accuracy of the result. (CO1)

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OR

- (b) A resistor R_1 has a potential difference of 25 V across its terminals and a current of 63 mA. The voltage is measured on a 30 V analog instrument with an accuracy of $\pm 5\%$ of full scale. The current is measured on a digital instrument with a ± 1 mA accuracy. Calculate the resistance of R_1 and specify its tolerance. (CO1)
3. (a) What is dynamic response ? Explain speed of response, response time, lag, fidelity and overshoot. (CO1 & CO2)

OR

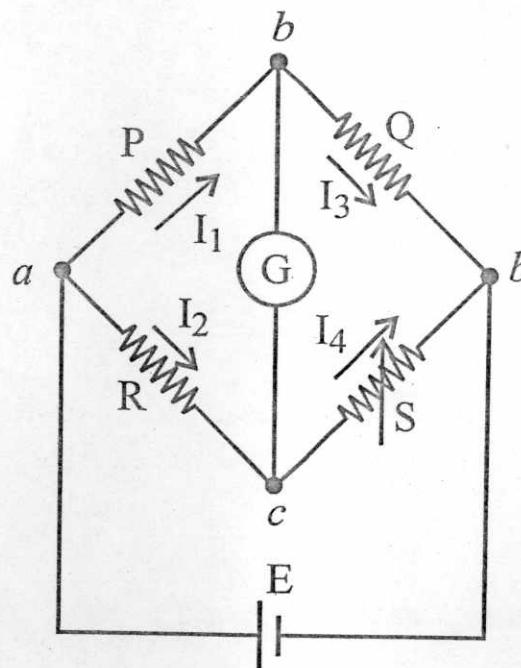
- (b) Explain the working principle of PMMC type instrument, moving iron type instrument and electrodynamicometer type instrument. Also, give the expression of turning torque generated by each of the above-mentioned instruments. (CO1 & CO2)
4. (a) An ammeter has a PMMC instrument with a coil resistance of 99 ohm and full-scale deflection current of 0.1 mA the shunt resistance of 1 ohm. Determine the total current passing through the ammeter at (i) full-scale deflection, (ii) 0.5 full-scale deflection, (iii) 0.25 full-scale deflection. (CO2 & CO3)

OR

- (b) Sketch the circuit diagram for a multirange voltmeter using (i) individual multiplier resistors and (ii) series-connected multiplier resistors. Explain the circuit operation in each case with suitable example. (CO2 & CO3)

(3)

5. (a) In the Wheatstone bridge of Fig., the values of resistances of various arms are $P = 1000 \text{ ohm}$, $Q = 100 \text{ ohm}$, $R = 2005 \text{ ohm}$ and $S = 200 \text{ ohm}$. The battery has an e.m.f. of 5 V and negligible internal resistance. The galvanometer has a current sensitivity of $10 \text{ mm}/\mu\text{A}$ and an internal resistance of 100 ohm . Calculate the deflection of galvanometer and the sensitivity of the bridge in terms of deflection per unit change in resistance. (CO3)



OR

- (b) Explain the Ammeter-Voltmeter method for measuring resistance with the help of a suitable example. (CO3)